

TECHNICAL GUIDE & COOKBOOK

yt-dlp Usage Guide

Practical Media Downloading, Quality Control & Transcoding

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1. Overview

`yt-dlp` is the industry-standard command-line media downloader, supporting thousands of video and audio sites. Under the hood, modern platforms (like YouTube) serve video and audio as separate adaptive streams (DASH). `yt-dlp` automatically downloads the best matching streams and merges them using `ffmpeg`.

💡 Key Principle: Dynamic Format Selection over Hardcoded IDs

Hardcoding static format ID numbers is unreliable because video platforms change or phase out format IDs across different videos. Use **dynamic resolution constraints** (e.g. `bv*[height<=720]+ba/b`) to ensure consistent, reliable downloads across any link.

2. Most Used Commands (The "Essentials")

2.1 Best Quality Video (Default Download)

Running `yt-dlp` without flags automatically selects the highest available video and audio quality, merging them into a single file:

TERMINAL — BEST QUALITY

```
yt-dlp "https://www.youtube.com/watch?v=wt4p2oalRY"
```

2.2 Inspecting Available Formats & Resolutions (-F)

Before downloading, use capital `-F` (or `--list-formats`) to "peek" at all available resolutions, frame rates, bitrates, audio tracks, and codecs offered by the target URL without downloading anything:

TERMINAL — LIST AVAILABLE RESOLUTIONS & FORMATS

```
yt-dlp -F "https://www.youtube.com/watch?v=wt4p2oalRY"
```

2.3 Download as MP4 (Maximum Player Compatibility)

High-resolution video streams often default to WebM or MKV formats. To ensure the final output is saved in an MP4 container for maximum compatibility with phones, TVs, and editing tools:

TERMINAL — FORCE MP4 CONTAINER

```
yt-dlp --merge-output-format mp4 "URL"
```

2.4 Audio Downloading: Direct Stream vs. Transcoding

There are two distinct methods to download audio depending on your requirements:

METHOD A: DIRECT STREAM DOWNLOAD (FASTEST / NO CPU OVERHEAD)

```
# Download best available raw audio stream directly (no video download)
```

```
yt-dlp -f "ba/b" "URL"
```

```
# Download native M4A/AAC stream directly
```

```
yt-dlp -f "ba[ext=m4a]/ba" "URL"
```

METHOD B: CONVERSION TO MP3 OR M4A (TRANSCODE WITH FFMPEG)

```
# Extract and convert to highest quality MP3
```

```
yt-dlp -x --audio-format mp3 --audio-quality 0 "URL"
```

```
# Extract and transcode to M4A
```

```
yt-dlp -x --audio-format m4a "URL"
```

⚡ Performance & Resource Tradeoff

Method A fetches *only* the audio track straight from the server, saving bandwidth and instant processing time. **Method B** (`-x`) downloads the container and invokes `ffmpeg` to transcode the audio, which uses **more CPU power, time, and network bandwidth**.

2.5 Controlling Save Location & Filenames

Use the `-P` flag to specify the output directory, and `-o` to customize file naming templates:

TERMINAL — PATH & OUTPUT FORMATTING

```
# Save to ~/Downloads folder
```

```
yt-dlp -P "~/Downloads" "URL"
```

```
# Custom filename template: Title [VideoID].ext
```

```
yt-dlp -P "~/Downloads" -o "%(title)s [%id)s].%(ext)s" "URL"
```

3. Dynamic Resolution & Quality Selection

When you want to limit resolution (e.g. to save bandwidth or disk space), use height constraint selectors. `yt-dlp` will pair the best video stream matching your height threshold with the best audio track.

720p HD Maximum

```
yt-dlp -f "bv*[height<=720]+ba/b" "URL"
```

360p Low Bandwidth

```
yt-dlp -f "bv*[height<=360]+ba/b" "URL"
```

1080p Full HD Maximum

```
yt-dlp -f "bv*[height<=1080]+ba/b" "URL"
```

4K Ultra HD (2160p)

```
yt-dlp -f "bv*[height<=2160]+ba/b" "URL"
```

3.1 Video-Only Streams (No Audio)

If you only require the visual track (e.g., background footage for video editing):

TERMINAL — VIDEO ONLY

```
yt-dlp -f "bv*[height<=720]" "URL"
```

3.2 Preferring Specific Video Containers

To explicitly request MP4 video streams paired with M4A audio streams:

TERMINAL — MP4 PREFERENCE SYNTAX

```
yt-dlp -f "bv*[ext=mp4]+ba[ext=m4a]/b[ext=mp4]" "URL"
```

3.3 Understanding Fallback Syntax (`/` Operator)

The forward slash (`/`) in `yt-dlp` format strings functions as a **left-to-right "OR / Fallback" operator**:

- `bv*[height<=720]+ba/b`: First tries to find a video stream $\leq$ 720p paired with best audio. If *no* video stream $\leq$ 720p exists, it falls back to `/b` (best available pre-merged single stream).
- `bv*[height<=720]+ba/bv*+ba/b`: First tries $\leq$ 720p + audio. If unavailable, falls back to best video of *any* height + audio before resorting to single streams.
- `bv*[ext=mp4]+ba[ext=m4a]/bv*+ba/b`: Prefers native MP4 video + M4A audio. If unavailable, falls back to any container format (WebM/MKV).

Essential Rule: Always Enclose Format Selectors in Quotes

Characters like `*`, `+`, and `?` are interpreted as wildcards by terminal shells (Bash/Zsh). Always wrap `-f` options and URLs in double quotes (`"..."`) to avoid syntax errors.

4. Pro-Tips for Power Users

4.1 Batch Downloading from a List (`-a`) & Archive Tracking

Create a plain text file (e.g. `list.txt`) with one URL per line. Combine `-a` with any format selector:

TERMINAL — BATCH DOWNLOADING RECIPES

```
# 1. Best Quality Video+Audio (Default)
yt-dlp -a list.txt -P "~/Downloads"

# 2. Capped at 720p HD Video+Audio (Saves bandwidth across many links)
yt-dlp -f "bv*[height<=720]+ba/b" -a list.txt -P "~/Downloads/720p"

# 3. Direct Native Audio Only (M4A/AAC — Fast & lightweight)
yt-dlp -f "ba[ext=m4a]/ba" -a list.txt -P "~/Music"

# 4. Growing URL List: Auto-skip previously downloaded items
yt-dlp --download-archive archive.txt -a list.txt -P "~/Downloads"
```

Managing Growing URL Lists with `--download-archive`

`yt-dlp` loads `list.txt` at launch. When continuously appending new URLs to a list over time, passing `--download-archive archive.txt` logs finished video IDs and automatically skips previously downloaded videos on future runs without wasting time or network requests.

4.2 Playlist & Subtitle Management

- Download Single Video Only (Ignore Playlist):** Add `--no-playlist`
- Download Specific Playlist Items:** `--playlist-items 1-5`
- Embed Subtitles & Metadata:** `--write-sub` `--sub-langs "en.*"` `--embed-sub` `--embed-thumbnail`

4.3 Maintenance & Troubleshooting

Video platforms frequently update their APIs. Keep `yt-dlp` up to date to prevent download failures:

TERMINAL — MAINTENANCE COMMANDS

```
# Update yt-dlp to latest release
yt-dlp -U

# Pass browser cookies if encountering HTTP 403 Forbidden errors
yt-dlp --cookies-from-browser chrome "URL"
```

5. Quick Reference Table

Goal / Workflow	Recommended Command Flag	Example Command
Inspect Formats & Quality	<code>yt-dlp -F "URL"</code>	<code>yt-dlp -F "https://..."</code>
Best Quality (Default)	<code>yt-dlp "URL"</code>	<code>yt-dlp "https://..."</code>
Max 720p HD	<code>-f "bv*[height<=720]+ba/b"</code>	<code>yt-dlp -f "bv*[height<=720]+ba/b" "URL"</code>
Max 1080p Full HD	<code>-f "bv*[height<=1080]+ba/b"</code>	<code>yt-dlp -f "bv*[height<=1080]+ba/b" "URL"</code>
Direct Audio Only (Fast)	<code>-f "ba/b"</code>	<code>yt-dlp -f "ba/b" "URL"</code>
Convert to MP3/M4A (FFmpeg)	<code>-x --audio-format mp3</code>	<code>yt-dlp -x --audio-format mp3 --audio-quality 0 "URL"</code>
Video Only (No Audio)	<code>-f "bv*[height<=720]"</code>	<code>yt-dlp -f "bv*[height<=720]" "URL"</code>
Batch Native Audio	<code>-f "ba[ext=m4a]/ba" -a list.txt</code>	<code>yt-dlp -f "ba[ext=m4a]/ba" -a list.txt -P "~/Music"</code>
Growing List (Archive)	<code>--download-archive archive.txt -a file</code>	<code>yt-dlp --download-archive archive.txt -a list.txt</code>
Download Directory	<code>-P "PATH"</code>	<code>yt-dlp -P "~/Downloads" "URL"</code>
Update Tool	<code>-U</code>	<code>yt-dlp -U</code>

6. Recommended Shell Aliases

Add these handy shortcuts to your `~/.bashrc` or `~/.zshrc` file for fast daily downloading:

```
BASH / ZSH CONFIG (~/.BASHRC)

# yt-dlp Shortcuts
alias ytdl='yt-dlp'
alias ytdlinfo='yt-dlp -F'
alias ytdl720='yt-dlp -f "bv*[height<=720]+ba/b"'
alias ytdl1080='yt-dlp -f "bv*[height<=1080]+ba/b"'
alias ytdlaudio='yt-dlp -f "ba/b"'
alias ytdlmp3='yt-dlp -x --audio-format mp3 --audio-quality 0'
```